STATE OF LIFE

parkrun

An experimental first look at the social and economic valuation of parkrun using the brand new 2021 HMT Green Book guidance and the WELLBY measure.

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1. Executive summary

This is a pioneering study that uses the brand new 2021 Treasury supplementary guidance on wellbeing valuation and the WELLBY measure. **The findings are eye opening and we invite scrutiny, discussion and more studies using physical activity and preventative health measures and the WELLBY.**

This new WELLBY approach could help policy makers better understand and quantify the cost effectiveness of physical activity as a health intervention vs the costs of NHS treatment for ill health.

- Parkrun launched as a one-off event in a London park in 2004. It's now a global health initiative that delivers weekly, free, community events in 2,200 locations across 24 countries.

- The new WELLBY is a unit of measurement of wellbeing impacts, which can be easily converted into a monetary value. It is based on a very simple and direct survey question - "All things considered, how satisfied are you with your life nowadays?" - 0 (not at all) to 10 (completely).

- This study applies the latest (2021) HM Treasury Green Book guidance and measures, including the WELLBY and the recommended valuation rate of £13,000/WELLBY, to assess the wellbeing impact attributable to parkrun and its equivalent monetary value. This approach led us to produce two estimates of the wellbeing impact and monetary value of parkrun in the UK. Both are remarkable.

- The first approach estimates the increased physical activity and volunteering as a result of parkrun and values these **using generic benefits of physical activity** and volunteering derived from UK longitudinal household surveys. Using this approach, the social return on investment (ROI) of parkrun UK is £30 for every £1.

- The second approach uses **parkrun's bespoke survey data** to estimate the specific wellbeing benefit of walking/running and volunteering at parkrun UK events. Using this approach, the social ROI is a massive £120 for each £1 invested.

- Sport England's broad social ROI study suggests sport has a return of £4 for every £1 spent. This report therefore suggests that parkrun is somewhere between 8 and 30 times more cost effective at producing a positive social ROI than most other sports.

- There is also some evidence, as with all sports, that the health and wellbeing benefits of participating in free, weekly parkruns in a local park are more beneficial for lower socio economic groups. This is in line with the UK Government's Levelling Up agenda and Sport England's new strategy 'Uniting the movement'.

The new WELLBY-based cost-benefit analysis enables a pioneering look at how physical activity based health and wellbeing interventions compare to the NHS in terms of cost effectiveness

- Particularly relevant is the fact that the new WELLBY can also be linked to the NHS measure for health improvements - the QALY[1] (Quality Adjusted Life Year), enabling a broad comparison of between the value and cost effectiveness of physical activity based preventative health and wellbeing programmes like parkrun as compared to what the NHS costs would be for similar outcomes.

- This wellbeing cost-effectiveness analysis and comparison is pioneering but it also suggests that parkrun as a health measure is much more cost-effective than the NHS in terms of its ability to generate social welfare in the population. This is largely because parkrun focuses on managing and preventing health problems in a non-medical environment rather than clinicians treating and curing these problems in hospitals, which involves significant expenses.

- This is not to suggest parkrun and other physical activity programmes and preventative measures are in any way a substitute for the NHS, but that their value as a cost reducing complement to the NHS seems positive and relevant. This focus on preventative health is clearly outlined in the 2019 Green Paper on the future of health prevention[2].

- There is enormous potential in this new measure of the WELLBY to determine what projects and programmes are most cost effective at improving the health of the nation; we hope this pioneering study kicks off more interest in what could be vital, thought provoking work in the physical activity and health sectors.

[1] https://en.wikipedia.org/wiki/Quality-adjusted_life_year

[2] https://www.gov.uk/government/consultations/advancing-our-health-prevention-in-the-2020s/advancing-our-health-prevention-in-the-2020s-consultation-document

2. Introduction

What is parkrun?

A UK-based charity, parkrun is a global health initiative that delivers a series of weekly, free, community events, designed to encourage engagement with physical activity and volunteering in a social environment.

Launched in October 2004 as a single event in Bushy Park, London, parkrun now supports weekly events in over 2,200 locations across 24 countries.

The charity's mission is to make the world healthier and happier through the provision of regular, community-led, socially-focussed physical activity events.

Parkrun events take place every Saturday morning in the form of 5km walks or runs for anyone aged four or over; there is also a 2km junior parkrun series for children aged four to fourteen. On average, each parkrun event is delivered by 20 volunteers, drawn from the local community and spanning a wide range of roles.

What is the WELLBY?

The WELLBY is a new, simpler measure of economic value that has recently gained traction in social value studies. It is based on a very simple and direct survey question - "All things considered, how satisfied are you with your life nowadays? - 0 (not at all satisfied) to 10 (completely satisfied)." 1 WELLBY is equal to one person moving 1 point on this 10-point scale for one year as a result of the policy/programme.

In 2021 the WELLBY was endorsed by the UK Treasury in its HM Treasury Green Book supplementary guidance on wellbeing. The HM Treasury Green Book is the UK government's handbook on policy appraisal i.e. the gold standard of UK policy evaluation. It states unequivocally that social value is based on the principles of welfare economics and includes all significant costs and benefits that affect the wellbeing of any member of the population, not just market effects.

Ways to convert impacts measured in WELLBY into equivalent monetary values have been further honed in recent years. They are presented in detail in the State of Life WELLBY Value Guide,

while this blog article summarises the basics. Particularly relevant is that the WELLBY is linked to the NHS measure for health improvements - the QALY[3] (Quality Adjusted Life Year).

The purpose of this study

With the release of the aforementioned supplementary guidance, wellbeing measured via self-reported life satisfaction (as described above) can now be officially used to gauge the social value of numerous programmes and interventions. State of Life has been doing this since its foundation as Jump Projects in 2016, particularly in the areas of volunteering, youth and sport.

The purpose of this work for parkrun is to apply the latest Green Book guidance and measures, including the new WELLBY measure, to quantify the impact and equivalent monetary value of parkrun on personal wellbeing in the UK. The study has two distinct parts that are used to check and complement each other.

Part 1 - using existing data to create a rough estimate of the value of parkrun
In one of State of Life's most important pieces of work (Faith, Hoops and Charity)[4], we established the wellbeing benefits of regular participation in sport and physical activity, attendance at religious services, and volunteering by applying robust econometric techniques to large data sets from multiple waves of UK longitudinal household surveys. These findings can be used to produce standard wellbeing values for getting one person to do physical activity at various frequency levels (weekly, monthly etc.) over the course of one year, and similarly for volunteering.

These standard values can be applied to parkrun participants (defined as walkers, runners and volunteers) to produce a rough estimate of the social value of the wellbeing benefits of parkrun (including both walking/running and volunteering at parkrun events). We elaborate on the details in the next section (Part 1).

Part 2 - using parkrun's own longitudinal data
We refine the estimates of Part 1 by using bespoke data on life satisfaction before and after parkrun participation from the parkrun UK Health and Wellbeing longitudinal survey to produce a figure that is more accurate and relevant to parkrun (Part 2).

The conclusion

The conclusions of both parts of this report are powerful. Parkrun is a lean organisation with a low cost base (no membership fees, grounds or club house to maintain). The weekly events are free to take part in and delivered by committed volunteers. Using both existing national data and

[3] https://en.wikipedia.org/wiki/Quality-adjusted_life_year
[4] https://www.stateoflife.org/our-work

parkrun's own data, we have found that participating in parkrun has significant positive impacts on health and wellbeing.

The result is that parkrun is an incredibly cost-effective way to improve health and wellbeing.

The return on investment for £1 spent is either £30 (in part 1) or £120 (in part 2). This is astonishing when anything over £1 is a positive return on investment and a return of £3 or £4 is broadly accepted as a successful return on investment.

Sport England's social return on investment study[5] suggests sport has a return of £4 for every £1 spent. This report suggests that parkrun is up to 30 times more cost effective than most other sports.

We also use the new WELLBY measure to benchmark the cost effectiveness of parkrun to the NHS. Parkrun seems to be between 6 to 25 times more cost effective than the NHS at generating wellbeing and this comparison does start to make clear the massive cost effectiveness potential of health and physical activity programmes. Especially those that remove barriers to event delivery and participation (free or low cost to set up and access) and are delivered at scale by a local community of volunteers, i.e. parkrun.

However, the above paragraph should not be interpreted as stating parkrun is better than the NHS. The two interventions fulfil different primary roles - prevention of health problems and improvement of health and wellbeing, versus active treatment of health problems (although it must be acknowledged that the NHS also has a preventative health role). The two primary interventions are most effective in combination, but it is more cost effective to allocate resources to prevention rather than treatment down the line.

Finally there is evidence, as with all sports, that the health and wellbeing benefits of participating in free, weekly parkruns in your local park can be more beneficial to lower socioeconomic groups.

At a time of post pandemic health and socioeconomic recovery, leveling up and a need to be smart with spending, parkrun looks to have an important role to play.

[5] https://www.sportengland.org/news/why-investing-physical-activity-great-our-health-and-our-nation

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Part 1. Rough estimate using existing sport and volunteering values

Faith, Hoops and Charity - the WELLBY in practice

The 2020 work Faith Hoops and Charity showed, using fixed effect regression analysis, that weekly sport is producing a 0.135 coefficient on life satisfaction on a 1-7 scale. The regression controlled for a range of demographic factors that strongly influence personal wellbeing - age, gender, ethnicity, religion, income, marital status and number of children, general health etc. Therefore we infer that a person who does weekly sport as a minimum is 0.135 points better off than a person of the same age, gender, health, income etc. who doesn't do any sport.

This means that any programme that gets one person (who otherwise would not have done any sport) to do sport weekly for one year will have a wellbeing impact of 0.225 WELLBY (because 0.135 on a 1-7 scale is linearly rescaled to 0.225 on the standard 0-10 life satisfaction scale).

Similar results have been produced for other frequencies of engaging in sport and physical activity, and likewise for volunteering (see Table 8 of the full technical paper). If we apply to these wellbeing coefficients the updated wellbeing valuation rates recommended by the latest Wellbeing Supplementary Guidance of the UK Government, we get the following values per person per year:

Table 1. Wellbeing valuation of physical activity and volunteering based on Faith, Hoops and Charity regression findings and latest UK government Wellbeing Supplementary Guidance

	Coefficient	Values based on 2021 wellbeing guidance update		
	(1 to 7 scale)	Lower bound	**Central**	Upper bound
Frequency of exercise:				
Never/almost never	0	£0	**£0**	£0
Once a year or less	0.014	£233	**£303**	£373
Several times a year	0.086	£1,433	**£1,863**	£2,293
At least once a month	0.094	£1,567	**£2,037**	£2,507
At least once a week	0.135	£2,250	**£2,925**	£3,600
Frequency of volunteering:				

Never/almost never	0	£0	**£0**	£0
Once a year or less	-0.042	-£700	**-£910**	-£1,120
Several times a year	0.038	£633	**£823**	£1,013
At least once a month	0.078	£1,300	**£1,690**	£2,080
At least once a week	0.095	£1,583	**£2,058**	£2,533

parkrun - the number of participants, counterfactual and additionality

The other 'piece of the puzzle' to get the total social value of parkrun is the number of people involved in any given year. We obtained from client records the number of people involved in parkrun UK events in 2018 and 2019[6] separated by the number of walks/runs completed / occasions volunteered. We grouped these as follows, to correspond to the five frequency categories analysed in 'Faith Hoops and Charity':

- 1 occasion - 'once a year or less'
- 2-11 occasions - 'several times a year'
- 12-51 occasions - 'at least once a month'
- 52 or more occasions - 'at least once a week'

The resulting numbers of unique participants and volunteers are provided below:

Table 2. Number of unique parkrun participants in 2018 and 2019

	2018			2019		
	2K and 5K	Just 5K	Just 2K	2K and 5K	Just 5K	Just 2K
Walking/Running						
Once a year or less	289,204	261,582	39,189	342,064	308,679	47,911
Several times a year	476,068	419,793	66,237	558,245	488,960	81,756
At least once a month	173,400	157,237	15,770	215,006	194,081	20,459
At least once a week	955	559	1	1,595	1,071	1
Total walking/running	939,627	839,171	121,197	1,116,910	992,791	150,127
Volunteering[7]						

[6] This also implies that the data is not affected by the negative impacts of the COVID-19 outbreak.
[7] People counted in the volunteering section may overlap with people in the running/walking section. However, since the benefits were proven to be additive, this does not affect the key findings.

Once a year or less	53,556	48,089	10,964	64,176	56,936	14,052
Several times a year	67,702	58,337	13,953	80,375	68,631	17,177
At least once a month	18,462	12,788	5,818	23,281	16,098	7,403
At least once a week	380	50	3	490	88	5
Total volunteering	140,100	119,264	30,738	168,322	141,753	38,637

Finally, there is one extra step that needs to be considered before we arrive at the final benefit figures. An evaluation of a policy or programme (such as parkrun) always needs to be considered against a well-defined **counterfactual scenario** - that is, what would have happened, all things considered, if the programme was not in place?

If parkrun did not exist, some of the people running/walking and volunteering at parkrun events would have run/walked and volunteered anyway. The goal is then to identify the proportion of the total population of parkrunners that would not have been involved in physical activity had it not been for parkrun, also known as the **additionality rate**. The true additionality rate is unknown (because it is the result of the comparison to a counterfactual scenario which is imaginary, and therefore no real data exists for comparison). Therefore additionality is provided by assumption - an educated guess using the maximum extent of relevant data provided by parkrun that might help us imagine the counterfactual scenario.

The below table documents the amount of physical activity per week that parkrunners were undertaking prior to starting parkrun. The question was asked at the moment of registration for parkrun. Based on this table, we can assume that all but 6% of parkrunners already had a sufficiently high physical activity level before they took up parkrun - enough to classify into the highest frequency category of physical activity in the Faith, Hoops and Charity paper. While waiting for better evidence, we apply an additionality rate of 6% to estimate parkrun's impact on people's participation in regular physical activity.

Table 3a. Prior physical activity levels

Pre-parkrun activity (5K, 2018)	Unique parkrunners	Share, %
Less than once per week	49,101	6.2%
About once per week	101,670	12.9%
About twice per week	192,163	24.4%
About three times per week	245,869	31.3%
Four or more times per week	197,262	25.1%

The parkrun Volunteer Survey 2019 shows that 41-43% of respondents had no experience of volunteering prior to getting involved with parkrun. Based on that, we assume an additionality rate for volunteering of 40%.

Table 3b. Prior volunteering levels

Before I volunteered at parkrun, in the previous 3 years:	March 19	June 19	Sept 19	Dec 19
I had **no experience** of volunteering in sport or community context	43.7%	42.0%	41.3%	41.7%
I had **some experience** of volunteering in sport or community context	37.5%	39.0%	40.1%	39.4%
I had **lots of experience** of volunteering in sport or community context	18.9%	19.0%	18.6%	18.9%

Alternative approach: wellbeing cost-effectiveness analysis

There is also an alternative method for assessing policies and programmes named Social Cost-Effectiveness Analysis.

Rather than trying to convert wellbeing impacts (WELLBYs) into pounds, this method ranks the policy alternatives in terms of their efficiency of achieving positive wellbeing impacts - the policy with the lowest £ spent per WELLBY generated ratio is deemed optimal (this can also be reported the other way round - the highest WELLBY generated per £ spent).

For example, for large NHS spending programmes designed to improve health and quality of life, the wellbeing cost-effectiveness ratio was estimated in several complex studies to be around £2,500 of spend per WELLBY.

This means that if parkrun had a corresponding cost / wellbeing benefit ratio lower than £2,500 per WELLBY, it would be more efficient (cost-effective) in producing personal wellbeing than the NHS. One should keep in mind that the ways through which parkrun and the NHS generate welfare are different, and that the best outcomes for the population may be achieved through a combination of preventing health problems occurring and effective treatment of health problems.

Costs for parkrun

These are taken from parkun's annual report[8] and consolidated financial statements for the financial year ending 31 January 2020. On page 32, it lists total expenditure in 2019 (1 February 2019 - 31 January 2020, to be exact) as £4,549,599 and in 2018 as £3,475,390 respectively.

[8] Note that these cost figures are for parkrun as a global operation and not just the UK so likely to understate the cost benefit in the UK only as reflected in this report.

A first estimate of a WELLBY-based value for parkrun

The inputs mentioned above (wellbeing value per person, number of walkers/runners (or volunteers respectively) and additionality rate) are multiplied together, producing the results below. To summarise what we have done:

- The frequency of participation has been linked to the relevant wellbeing impact from Faith Hoops and Charity (table 1 above)
- We have then applied (and converted from the 1-7 scale) to this coefficient the 2021 value of a WELLBY at £13,000 for a 1 point increase in Life Satisfaction on a 0-10 scale
- Finally we have applied an additionality rate of 40% (or what would also be a deadweight/discount rate of 60% to account for the counterfactual) for volunteering and an additionality rate of 6% (deadweight of 94%) for walking/running

Table 4. Wellbeing benefits of parkrun

	2018			2019		
	2K and 5K	Just 5K	Just 2K	2K and 5K	Just 5K	Just 2K
Walking/Running:						
Once a year or less	£5,263,513	£4,760,792	£713,240	£6,225,565	£5,617,958	£871,980
Several times a year	£53,224,402	£46,932,857	£7,405,297	£62,411,791	£54,665,728	£9,140,321
At least once a month	£21,189,480	£19,214,361	£1,927,094	£26,273,733	£23,716,698	£2,500,090
At least once a week	£167,603	£98,105	£176	£279,923	£187,961	£176
Total walking/running	**£79,844,998**	**£71,006,116**	**£10,045,806**	**£95,191,012**	**£84,188,345**	**£12,512,566**
Additionality	6%	6%	6%	6%	6%	6%
Volunteering:						
Several times a year	£22,296,525	£19,212,319	£4,595,188	£26,470,167	£22,602,476	£5,656,959
At least once a month	£12,480,312	£8,644,688	£3,932,968	£15,737,956	£10,882,248	£5,004,428
At least once a week	£312,867	£41,167	£2,470	£403,433	£72,453	£4,117
Total volunteering	**£35,089,704**	**£27,898,173**	**£8,530,626**	**£42,611,556**	**£33,557,177**	**£10,665,503**
Additionality	40%	40%	40%	40%	40%	40%
Grand total	**£114,934,702**	**£98,904,289**	**£18,576,432**	**£137,802,568**	**£117,745,522**	**£23,178,070**

Thus the total yearly social value of walking and running at parkrun events was £80 million in 2018 and £95 million in 2019. The value of volunteering was £35 million in 2018 and £43 million in

2019 respectively, for a total wellbeing value of parkrun of £115m in 2018 and £138m in 2019. The social value in 2019 increased around 20% compared to 2018 levels.

Cost-Benefit Analysis, Cost-Effectiveness Analysis and final metrics

Considered alongside the costs provided in parkun's latest annual report, parkun delivered a net benefit of £111 million in 2018 and £133 million in 2019. **For every £1 invested, the social value returned was over £33 in 2018 and over £30 in 2019**. Wellbeing cost-effectiveness analysis shows that parkrun needs to spend around £400 for every WELLBY generated among the population of its participants, which makes it over 6 times cheaper than the NHS as a producer of social welfare in the UK[9].

Table 5. Social CBA and Wellbeing CEA of parkrun

	2018	2019
Social Cost-Benefit Analysis:		
Benefits of walking and running	£79,844,998	£95,191,012
Benefits of volunteering	£35,089,704	£42,611,556
Total benefits	£114,934,702	£137,802,568
Total costs	£3,475,390	£4,549,499
Net benefit	**£111,459,312**	**£133,253,069**
Benefit-cost ratio	**33.07**	**30.29**
Wellbeing Cost-Effectiveness Analysis:		
Impact of walking and running in WELLBYs	6,142	7,322
Impact of volunteering in WELLBYs	2,699	3,278
Total wellbeing impact in WELLBYs	8,841	10,600
Wellbeing cost-effectiveness ratio: £/WELLBY	**£393**	**£429**

When we look at other sports and activities on a CBA / SROI basis we can see that parkrun is extraordinarily good value.

Sport England's broad social return on investment study[10] suggests sport has a return of £4 for every £1 spent and recent studies by State of Life for other sports are of the same magnitude. A

[9] Frijters and Krekel (2021) calculate, based on a study by Claxton et al. (2015), that the NHS spends around £2,500 for each WELLBY generated by its health-related interventions.
[10] https://www.sportengland.org/news/why-investing-physical-activity-great-our-health-and-our-nation

return of £30 or more for every £1 invested suggests parkrun is 8 times more effective at generating health and wellbeing benefits than most other sports. It would be interesting to start to look at the costs and WELLBY value of more sports alongside parkrun.

The cost effectiveness of parkrun might, in part, be attributed to a lower cost base - there is no clubhouse or grounds that parkrun has to maintain. But at the same time the sheer scale of parkrun's model and participation is what produces the big return on investment.

The WELLBY cost effectiveness measure is similarly impressive. The NHS production cost of a WELLBY is around £2,500 while parkrun is around £400. And again this can be explained with the enormous staff and facility costs of the NHS compared to parkrun's free access to parks and volunteer delivery.

It is important to stress we are not comparing like for like with parkrun to the NHS, but provide this as an illustration of how cost-effective free, accessible preventative and general health improvement interventions like parkrun could be in generating health to the UK. This is especially relevant when compared to the high costs of treating ill health born by the NHS.

Part 2. Using bespoke parkrun data

The above estimates are a rough approximation of the wellbeing benefits and value of parkrun. To be more accurate, it is best to try to measure the positive change in wellbeing that is specifically attributed to parkrun participation, rather than to general engagement in sport and physical activity or volunteering.

The data

Fortunately, in 2018 parkrun commissioned UK Health and Wellbeing Surveys that can be used for this purpose. There is both a cross-sectional survey and a longitudinal survey, which ask similar questions but have some important differences in the sampling design.

The cross-sectional survey (October - November 2018) is addressed to everyone who registered with parkrun and therefore captures respondents at various points in their history of parkrun participation. The main advantage of this survey is its sample size - over 60,000 respondents. Its main drawback is the lack of a suitable control group for parkrun participants.

Although the sample does contain ca. 1,700 respondents who registered but have not participated, we cannot rule out that their participation was prevented by conditions that make them fundamentally different from the remaining sample - health problems, lower socioeconomic status etc. The respective differences in the demographics are shown in Appendix Table A2.

The longitudinal survey is addressed to a specially selected sample of newly registered parkrun participants, and is conducted in 2 waves. The first wave (January - February 2019) is conducted just a few days after the respondent registered for parkrun (between 1 and 12 days, to be exact - by which time some respondents have already managed to participate once while others have not participated yet). The second wave is conducted 6 months after the first interview for each respondent (+- a few days). This survey is conducted on a much smaller sample - 567 individuals have completed both waves, for a total of 1134 responses (the remaining individuals, who do not have both a baseline and a follow-up response, are discarded). However, the benefit of observing the same individuals before and after exposure to parkrun considerably outweighs that, as it allows to negate any kind of selection bias (such as parkrun participants being more affluent or happier to begin with) by comparing like for like.

Therefore we use the longitudinal survey as the primary source for our analysis and use the cross-sectional survey for triangulation and to produce further input statistics that will be required for aggregation to calculate the total wellbeing value of parkrun (such as the average number of walks/runs and volunteering occasions per year for parkrun participants).

Methodology

We use multivariate regression analysis to reveal the association between life satisfaction and parkrun participation. This is based on the following model equations:

$LS_{it} = \alpha + \gamma_{t=2} + \sum x_{jit}\beta_j$ (1) and

$LS_{it} = \alpha + \delta Nruns_{it} + \theta Nvol_{it} + \sum x_{jit}\beta_j$ (2)

Where LS is individual i's life satisfaction on a scale of 0-10 at time t (t=1 for the baseline and t=2 for the 6-month follow-up).

x_j is a series of control variables that may be alternatively responsible for respondents' differing levels of life satisfaction, including employment, ethnicity, disability, participation type (runner/walker only vs. runner/walker and volunteer[11]), gender, age (simple and squared), physical activity at registration (ranging from 'less than once per week' to 'four or more times per week'), club affiliation and local area deprivation (IMD quartile).

We perform both pooled OLS and fixed effects regressions for these model setups. We give priority to the fixed effects results, because they account for time-constant individual-specific unobservable differences in wellbeing[12] . Note that most of the control variables (all except employment and age) are naturally omitted from the model setup when running fixed effects, because they do not change from baseline to follow-up or because they are only asked at baseline.

In equation (1), the coefficient of the follow-up survey wave $\gamma_{t=2}$ will show the average change in life satisfaction among our sample respondents after being exposed to parkrun for 6 months, holding constant all other factors mentioned above. However, the degree of involvement with parkrun varies strongly, and therefore we also employ a model with interaction terms to disaggregate the wellbeing change by participation type (walker/runner and volunteer vs. walker/runner only). Finally, equation (2) is able to better distinguish between various intensities of involvement by estimating a change for every run/walk instance and every volunteering occasion.

[11] Nobody in the longitudinal survey sample was a volunteer only.
[12] For a full explanation of these types of analysis see the papers 'Happy Days' and 'Faith, Hoops and Charity' on the State of Life website - https://www.stateoflife.org/our-work

There are some modifications of the model that we include for triangulation or as complementary findings. Firstly, we also include health as a control variable (self-reported on a 0-100 scale, with 100 being the best possible health and 0 - the worst possible health) because it is one of the most important observable determinants of wellbeing. However, we believe that in this study, health is also a channel through which the positive wellbeing effect of parkrun occurs: it is very likely that doing parkrun improves your health, which increases your life satisfaction as a result. Including health as a control variable removes this partial effect and therefore may understate the true effect. To be on the conservative side, we present the results from the model that controls for health in order to exclude any bias from health improvements that occurred from reasons not related to parkrun. We must stress that to control for and effectively remove health from what is fairly obviously an activity to benefit health is very likely to underestimate one of the key benefits of parkrun.

Secondly, we also consider a range of other outcomes instead of life satisfaction. These are happiness, various measures of physical activity, health (using the subjective 0-100 scale and also the EQ5D-5L scale), mental wellbeing (the short WEMWBS), and medical service use in the last 12 months (GP visits, days in hospital etc). While life satisfaction is the only outcome used for wellbeing valuation, these other outcomes are important to paint a broader picture of the benefits that parkrun participation offers, and reduced medical services use also bears an important cost saving to the government that can be quantified in future research.

Finally, we also analyse the cross-sectional parkrun Health and Wellbeing Survey using a similar model setup - although the participation variables are defined somewhat differently in the cross-sectional survey and only OLS is used (fixed effects models can only be employed with longitudinal data). Because of this the cross-sectional survey results are more likely to be biased by self-selection and are therefore not presented as main findings (we could put these in an Appendix?)

The wellbeing benefits of parkrun - results

Table 6. Wellbeing regression results - longitudinal survey

Outcome	6 months later	6 months later, walking/running and volunteering	6 months later, walking/running only	Per walk/run instance	Per volunteering occasion
Life satisfaction (0-10)	**0.273*****	**0.501*****	**0.234*****	**0.028*****	**0.015**
Happiness (0-10)	0.151*	0.298	0.127	0.016*	-0.016

Activity days/week (0-4)[13]	0.192***	0.237*	0.184***	0.023***	-0.003
Health (0-100)	3.802***	5.946***	3.453***	0.358***	-0.077
SWEMWBS (7-35)	0.466**	1.055**	0.369*	0.049**	0.055
GP visits (last 12 months)	-0.253**	0.013	-0.297**	-0.019**	0.008
Hospital admissions (12 m)	-0.037*	-0.103*	-0.027	-0.002	0.005
Days as an inpatient (12 m)	-0.100***	-0.263**	-0.076**	-0.006*	0.009
Minutes[14] of ph. act. / week	75.634**	215.716**	48.494	4.097	44.780***

Notes: ***p<0.01, **p<0.05, *p<0.1. Fixed effects regression models used with heteroskedasticity robust standard errors and with control variables listed in the methodology section. Full regression results not shown for brevity.

The other outcomes analysed - the sum of the EQ5D scores across the 5 domains, days of 30+ minutes of physical activity in the last week (0-7), days lost to sickness, times in contact with specialist from the NHS and outside the NHS, and times been to A&E (all in the last 12 months) did not exhibit almost any statistically significant associations with the parkrun involvement variables[15].

When we also include health as a control variable, there is still a positive and statistically significant association between participation in parkrun and the key wellbeing outcomes - life satisfaction, physical activity days/week (0-4), reduced GP visits, hospital and inpatient stays - albeit smaller in magnitude for life satisfaction and physical activity. This suggests that health improvements may be responsible for part of the wellbeing uplift associated with parkrun participation, but there are also other factors at work. Other outcomes (happiness, EQ5D, the WEMWBS, other physical activity measures) have a statistically insignificant association with involvement in parkrun. Results are shown in Table 6b below for comparison.

Table 6b. Wellbeing regression results - longitudinal survey, with health (0-100) control

Outcome	6 months later	6 months later, walking/running and volunteering	6 months later, walking/running only	Per walk/run instance	Per volunteering occasion

[13] Less than once a week coded as 0, four or more times a week coded as 4.
[14] Moderate activity equivalent minutes as per the IPAQ definition
[15] The only exceptions are: times been to A&E: -0.027*** per volunteering occasion, which does not have a very solid theoretical grounding, times in contact with specialist outside the NHS: -0.018* per run, and EQ5D sum: -0.227* for 6-months involvement, only those who walk/run and also volunteer.

Life satisfaction (0-10)	**0.176*** **	**0.335***	**0.151** **	**0.019*** **	**0.016**
Happiness (0-10)	0.010	0.078	-0.001	0.002	-0.011
Activity days/week (0-4)[16]	0.148***	0.216*	0.138**	0.019***	0.003
SWEMWBS (7-35)	0.193	0.623	0.126	0.024	0.060
GP visits (last 12 months)	-0.237*	0.043	-0.279**	-0.017**	0.008
Hospital admissions (12 m)	-0.040**	-0.107**	-0.030	-0.003	0.005
Days as an inpatient (12 m)	-0.107***	-0.274**	-0.083**	-0.007*	0.009
Minutes[17] of ph. act. / week	54.979	166.058	35.916	1.313	41.921**

Notes: ***p<0.01, **p<0.05, *p<0.1. Fixed effects regression models used with heteroskedasticity robust standard errors and with control variables listed in the methodology section + subjective self-reported health on a 0-100 scale. Full regression results not shown for brevity.

The cross-sectional survey (which we use to double check against the more robust longitudinal findings above) confirms the positive association between life satisfaction and parkrun participation, where the coefficient is 0.598*** for those who walk/run and volunteer, 0.503*** for those who only walk/run and 0.530*** for those who only volunteer, all as opposed to those who registered but never participated. Other parkrun involvement variables - years registered with parkrun, average number of walks/runs per year, total number of walks/runs, walks/runs in the last year, and total volunteering occasions - are also positively associated. The same is true for other health and wellbeing outcomes - happiness, various physical activity measures, EQ5D and subjective health, and reduced GP visits. Generally the regression coefficients are higher in magnitude than in the longitudinal data, but this is most likely because the subgroup of respondents who registered but never participated comes from a more disadvantaged background. This reinforces our use of the longitudinal data for the main results.

Clearly however, all the model variations above provide evidence that participating in parkun, whether as a walker/runner or as a volunteer or both, is associated with an increase in health and wellbeing. The exact effect size is harder to pin down and depends on the selected sample and model. Nonetheless, with the data provided we are able to use fixed effects regression analysis, which is considered a robust methodology according to most standards of evidence and considered sufficient for policy relevant analysis in the Magenta and Green Books. We therefore have sufficient confidence in the resulting estimates to be able to proceed with wellbeing valuation.

[16] Less than once a week coded as 0, four or more times a week coded as 4.
[17] Moderate activity equivalent minutes as per the IPAQ definition

Levelling up - the benefits of parkrun by deprivation group

One of the common findings in sport and physical activity is that there is a bias in the participants to higher socioeconomic groups. And parkrun is no different. State of Life's work in sport[18] has also shown that the benefits of participation on health and wellbeing are greater for lower socioeconomic groups.

We conducted analysis to see whether the conclusion from our previous work regarding higher wellbeing benefits for low socioeconomic groups is replicable for parkrun.

This is done by introducing a slight modification to our main regression models from Part 2, by including an interaction term between the parkrun participation variable and local area deprivation groups, represented by the quartile of the English Index of Multiple Deprivation (IMD). This estimates separate changes in wellbeing associated with parkrun for each of the four quartiles and is presented in Table 11 below:

Table 11. Life satisfaction regression results with health control disaggregated by deprivation

Subgroup	6 months later	6 months later, walking/running and volunteering	6 months later, walking/running only	Per walk/run instance	Per volunteering occasion
All (aggregate model)	0.176***	0.335*	0.151**	0.019***	0.016
1st quartile (most deprived)	**0.310***	**0.300**	**0.313***	**0.019**	**0.014**
2nd quartile	0.130	0.097	0.141	0.021*	-0.018
3rd quartile	0.169*	0.685**	0.075	0.014	0.035
4th quartile (least deprived)	0.168	0.149	0.179	0.024**	0.026

Notes: ***p<0.01, **p<0.05, *p<0.1. Fixed effects regression models used with heteroskedasticity robust standard errors and with control variables listed in the methodology section + subjective self-reported health on a 0-100 scale. Full regression results not shown for brevity.

In conclusion, the average benefits of parkrun involvement for 6 months are higher for the most deprived subgroup - almost twice as much as the average. The same is true for walkers/runners only, but not for those who walk/run and volunteer. In terms of the benefits per occasion, for volunteering we observe the opposite - higher associations for the less deprived groups, whereas for walking/running occasions the association is more or less constant across the board.

[18] https://www.stateoflife.org/our-work

Valuation and Social CBA using the 2021 Wellbeing Guidance

We now proceed to transform our key results - the association between life satisfaction and parkrun involvement derived from fixed effects regressions using the longitudinal survey - into equivalent monetary values using the rates recommended by the newly released 2021 Wellbeing Supplementary Guidance to the Green Book, as mentioned in the Introduction.

Table 7 below provides the respective wellbeing values per person of parkrun participation. Being involved with parkrun for 6 months is worth £2,300 on average in social value. This goes up to £4,400 if the involvement includes both walking/running and volunteering, and down to £2,000 for walking/running only. The wellbeing benefit of every extra walk/run can be quantified as £247, while the benefit of every extra volunteering occasion is £208.

Table 7. Wellbeing values per person of parkrun participation (unadjusted)

	Coefficient	Values based on 2021 wellbeing guidance		
	(0 to 10 scale)	Lower bound	**Central**	Upper bound
6 months parkrun involvement - average	0.176	£1,760	**£2,288**	£2,816
6 m. - walking/running and volunteering	0.335	£3,350	**£4,355**	£5,360
6 months - walking/running only	0.151	£1,510	**£1,963**	£2,416
per walk/run instance	0.019	£190	**£247**	£304
per volunteering occasion	0.016	£160	**£208**	£256

These results allow three different avenues to aggregate and estimate the total wellbeing value of parkrun in a year. They are highlighted by different background colours. One applies the average impact of parkrun involvement to all parkrun participants (both walkers/runners and volunteers). The second one disaggregates between walkers/runners/volunteers and walkers/runners only, and adds them back up together at the end. The third avenue is based not on the number of unique participants, but on the number of unique walking/running and volunteering occasions (added up together at the end).

The corresponding input quantities (number of participants and participation occasions respectively) are presented in the upper part of Table 8 below. The number of all participants is the sum of the number of unique walkers/runners and the number of unique volunteers - cf. Table 2. Note that there may be double counting as there is no certainty whether the unique walkers/runners and unique volunteers are also unique across the two groups. The next two rows use the proportions of walkers/runners/volunteers and walkers/runners only revealed by the cross-sectional parkrun survey - 35.1% and 60.9% respectively. Finally, the number of walks/runs and volunteering occasions are summed up directly from parkrun administrative records.

By multiplying these quantities with the values per person / per occasion from Table 7, we obtain the values in the middle section, which are then added as required to arrive at the total (unadjusted) yearly wellbeing values of parkrun for 2018 and 2019 using the three approaches, presented in the lower end of the table.

Table 8. Total unadjusted wellbeing values of parkrun

	2018			2019		
	2K and 5K	Just 5K	Just 2K	2K and 5K	Just 5K	Just 2K
	Number of participants					
Participants - all	1,079,727	958,435	151,935	1,285,232	1,134,544	188,764
Walk/run and volunt.	378,984	336,411	53,329	451,116	398,225	66,256
Walking/running only	657,554	583,687	92,528	782,706	690,937	114,957
Number of walks/runs	6,959,885	6,277,839	682,046	8,590,393	7,714,563	875,830
Volunt. occasions	787,289	581,166	206,123	974,255	711,777	262,478
	Wellbeing values					
Participants - all	£2,470,415,376	£2,192,899,280	£347,627,280	£2,940,610,816	£2,595,836,672	£431,892,032
Walk/run and volunt.	£1,650,476,091	£1,465,068,533	£232,248,601	£1,964,612,061	£1,734,269,631	£288,545,594
Walking/running only	£1,290,777,998	£1,145,777,414	£181,633,279	£1,536,452,443	£1,356,309,912	£225,661,133
Number of walks/runs	£1,719,091,595	£1,550,626,233	£168,465,362	£2,121,827,071	£1,905,497,061	£216,330,010
Volunt. occasions	£163,756,112	£120,882,528	£42,873,584	£202,645,040	£148,049,616	£54,595,424
	Total wellbeing values of parkrun					
Approach 1	£2,470,415,376	£2,192,899,280	£347,627,280	£2,940,610,816	£2,595,836,672	£431,892,032
Approach 2	£2,941,254,088	£2,610,845,947	£413,881,879	£3,501,064,505	£3,090,579,543	£514,206,727
Approach 3	**£1,882,847,707**	£1,671,508,761	£211,338,946	**£2,324,472,111**	£2,053,546,677	£270,925,434

However, we lend the highest level of credibility to approach 3, based on the individual walking/running and volunteering occasions. Firstly, this is because this approach better takes into account the varying frequencies of participation across individuals, rather than the first two approaches, which derive an average benefit for individuals in the longitudinal survey sample. Note that the average participation frequency in the longitudinal survey sample may not coincide with the population average participation frequency - in fact, parkrun administrative data shows an average of 7.4 walking/running occasions per each unique participant in 2018 and 7.7 in 2019, whereas the longitudinal average number of walks/runs completed at follow up is 8.87 vs. 0.51 at baseline.

Also, the proportion of walkers/runners/volunteers and walkers/runners only may not be accurately reflected in the sample - these are 14.3% and 85.7% respectively in the longitudinal survey vs. 35.1% and 60.9% in the much larger cross-sectional survey. Also the longitudinal survey has no respondents that are volunteers only or have not participated in parkrun. Hence both the total effect and its split between walkers/runners/volunteers and walkers/runners only may be distorted.

Finally, using walking/running and volunteering occasions as the counting unit bears the lowest risk of double counting - this is because these occasions are always unique in parkrun records, whereas a unique individual may be repeatedly counted in different activities (e.g. walking/running and volunteering, 5K and 2K events).

Adjustments to the wellbeing value of parkrun

Note that the values above are unadjusted - they do not include any deadweight or additionality assumptions. This is because our wellbeing analysis is specific to parkrun, and our control group consists of the same individuals when they were just barely beginning to participate, but may have still been well involved in other sports, physical activity and volunteering (in fact the survey data on prior physical activity confirms a part of this assumption). Prior physical activity levels are not a factor that can cause bias in this situation - they are controlled for in the regression equation.

However, one may still claim that this situation does not reflect the counterfactual fully accurately, and that in the absence of parkrun, people with the demographic profile that is likely to engage in parkrun would have also participated in a sizable proportion in similar activities with similar benefits (In the parkrun COVID impacts survey for England (October 2020), 38% reported they were now active on fewer days per week than they were pre-pandemic, with 22% feeling they were now more active. This means that 62% did not experience a reduction in activity levels even though parkrun was cancelled). Nonetheless, the size of this proportion remains the object of pure speculation.

Another very important thing to note is that the duration of the wellbeing effect remains unknown - because there was only one follow-up wave 6 months after registration, we cannot say whether this effect intensifies or drops off after a longer (or shorter) period of involvement, especially taking into account seasonal variations in participation observed in parkrun records. In particular for approach 3 - which we chose to generate our main result - the assumption is that every additional walk/run gives you a wellbeing uplift that lasts for one year, which might be too generous of an assumption.

To account for the two points above, we apply an adjustment and additionality factor of 25% to our final result. This implies that we assume the benefits of each extra walk/run either a) last for about 3 months on average or b) last about 6 months on average but only 50% of them are additional would not be replaced by similar activities in the counterfactual scenario where parkrun was unavailable.

As this adjustment factor crucially influences our final value, we also perform sensitivity analysis to illustrate how varying this parameter affects the final estimate for the social value of parkrun. This is shown in Table 9 below.

Table 9. The value of parkrun, additionality and deadweight

Additionality	Deadweight equivalent	Value of parkrun in 2018	Value of parkrun in 2019
100%	0%	£1,882,847,707	£2,324,472,111
80%	20%	£1,506,278,166	£1,859,577,689
60%	40%	£1,129,708,624	£1,394,683,267
50%	50%	£941,423,854	£1,162,236,056
33%	67%	£621,339,743	£767,075,797
25%	**75%**	**£470,711,927**	**£581,118,028**
20%	80%	£376,569,541	£464,894,422
10%	90%	£188,284,771	£232,447,211

Our central estimate for the value of the wellbeing benefits of parkrun (including walking/running and volunteering) is around £471 million in 2018 and £581 million in 2019.

The key message to take away is that when investigating the benefits of parkrun per se rather than its contribution to physical activity and volunteering more generally, the estimate of its wellbeing value may be 4 (with a 25% additionality adjustment) to 20 times higher (if no adjustment is used).

This invites the conclusion that parkrun is an activity that is positive for our wellbeing and extremely efficient in delivering social value for money, partly because of its low costs. This is illustrated in Table 10 below.

Table 10. Social CBA and Wellbeing CEA of parkrun using bespoke data

	2018	2019
Social Cost-Benefit Analysis:		

Total benefits	£470,711,927	£581,118,028
Total costs	£3,475,390	£4,549,499
Net benefit	**£467,236,537**	**£576,568,529**
Benefit-cost ratio	**135.44**	**127.73**
Wellbeing Cost-Effectiveness Analysis:		
Total wellbeing impact in WELLBYs	36,209	44,701
Wellbeing cost-effectiveness ratio: £/WELLBY	**£95.98**	**£101.78**

Thus, **parkrun produces a value of over £120 in wellbeing benefits for every £1 spent** - an astounding ratio given than a figure of £3 or £4 is broadly acceptable as a return in policy circles. Indeed,Sport England's social return on investment study[19] suggests sport has a return of £4 for every £1 spent.

This report therefore suggests that parkrun is up to 30 times more cost effective than most other sports and illustrates the power of accessible, free, grassroots sport and physical activity initiatives to make a difference in people's lives. Even if we were to assume an additionality rate of only 10% (rather than the 25% we use above) , the SROI / BCR would still be above 50 to 1 - a very high value. And the wellbeing cost-effectiveness analysis shows that parkrun is about 25 times more cost-effective in generating wellbeing improvements in the population than the NHS, which has an average cost per WELLBY of about £2,500 compared to parkrun cost per WELLBY of around £100.

We believe that the key reason why parkrun is so effective at delivering social value is not so much the benefits, which are significant but of a similar order to those of other widely practised sports we analysed in our previous work. It is rather the very low costs that make parkrun stand out - no club or membership fees, no participation fees, no facilities, kit or equipment required - parkrun is free for everyone. This makes parkrun much more cost-effective compared to traditional sports. And this also means that parkrun is a much more affordable, grassroots, community-level, health and wellbeing intervention than, say, the NHS or most traditional sports.

To make a community better off without spending a great deal of money it would seem that community and volunteer-led activity that is free at the point of use is a very powerful and cost effective health improving initiative.

[19] https://www.sportengland.org/news/why-investing-physical-activity-great-our-health-and-our-nation

Appendix of tables

Table A1. Demographics of the longitudinal survey

Follow-up vs. baseline	Baseline	Follow-up	Total
Sample Size	567	567	1134
Life satisfaction	7.58 (558)	7.84 (567)	7.71 (1125)
Happiness	7.61 (558)	7.76 (567)	7.68 (1125)
Single activity question 1-7 (integer)	3.39 (558)	3.34 (567)	3.36 (1125)
Eq5d - walking (1-best, 5-worst)	1.11 (556)	1.13 (567)	1.12 (1123)
Eq5d - washing and dressing	1.02 (555)	1.02 (567)	1.02 (1122)
Eq5d - usual activities	1.12 (556)	1.13 (566)	1.13 (1122)
Eq5d - pain and discomfort	1.49 (553)	1.51 (566)	1.50 (1119)
Eq5d - anxiety and depression	1.49 (555)	1.46 (566)	1.48 (1121)
Sum of EQ5D scores, 5 - best, 25 - worst	6.22 (555)	6.25 (566)	6.23 (1121)
Health /100	78.09 (554)	81.57 (567)	79.85 (1121)
Times in contact with GP	2.13 (539)	1.93 (560)	2.03 (1099)
Times in contact with NHS specialist	1.43 (500)	1.34 (540)	1.38 (1040)
Times in contact with specialist outside NHS	0.79 (458)	0.71 (512)	0.74 (970)
Times admitted to hospital	0.10 (455)	0.09 (512)	0.09 (967)
Times been to A&E	0.14 (450)	0.16 (511)	0.15 (961)
Days lost to sickness	5.69 (441)	6.58 (479)	6.15 (920)
Days as an inpatient	0.20 (455)	0.14 (514)	0.17 (969)
Days between survey and first run	12.86 (486)	. (0)	12.86 (486)
I've been feeling optimistic about the future	3.78 (543)	3.85 (561)	3.82 (1104)
I've been feeling useful	3.81 (543)	3.93 (557)	3.87 (1100)
I've been feeling relaxed	3.57 (544)	3.66 (562)	3.61 (1106)
I've been dealing with problems well	3.87 (544)	3.98 (561)	3.93 (1105)
I've been thinking clearly	4.09 (544)	4.15 (565)	4.12 (1109)
I've been feeling close to other people	3.83 (544)	3.91 (560)	3.87 (1104)
I've been able to make up my own mind about things	4.26 (545)	4.26 (563)	4.26 (1108)
Short WEMWBS total score, best - 35, worst - 7	27.19 (543)	27.54 (558)	27.37 (1101)
Vigorous minutes per week	124.97 (567)	124.42 (567)	124.69 (1134)
Moderate minutes per week	119.13 (567)	131.79 (567)	125.46 (1134)
Walking minutes per week	255.05 (567)	290.92 (468)	271.27 (1035)
Vigorous MET-mins /week	387.46 (469)	414.06 (394)	399.60 (863)
Moderate MET-mins /week	183.16 (469)	221.73 (394)	200.77 (863)
Walking MET-mins /week	185.66 (469)	202.62 (394)	193.40 (863)
TOTAL MET-mins /week	756.28 (469)	838.41 (394)	793.78 (863)
Number of parkruns completed	0.51 (486)	8.87 (499)	4.74 (985)
Volunteer Occasions	0.00 (486)	0.82 (499)	0.42 (985)

IMD %	0.61 (492)	0.61 (492)	0.61 (984)
Age (years)	49.91 (555)	50.52 (559)	50.21 (1114)
Age at registration (years)	50.01 (490)	50.04 (491)	50.03 (981)
Gender			
F	55.4%	55.4%	55.4%
M	44.6%	44.6%	44.6%
Employment			
Employed, but currently on maternity/paternity leave	0.4%	0.2%	0.3%
Employed, but currently on sick leave	1.1%	0.9%	1.0%
Full-time home maker	2.1%	2.3%	2.2%
Full-time paid employment	47.4%	47.8%	47.6%
Fully retired	17.1%	17.8%	17.5%
I'd rather not say	1.4%	0.5%	1.0%
Other	2.1%	1.4%	1.8%
Part-time paid employment	11.8%	13.9%	12.9%
Retired, but still in paid employment	3.2%	2.6%	2.9%
Self-employed	9.0%	8.5%	8.7%
Student	3.2%	2.1%	2.6%
Unemployed and not working	1.2%	1.9%	1.6%
Ethnicity			
African	0.5%	0.5%	0.5%
Bangladeshi	0.4%	0.4%	0.4%
Caribbean	0.5%	0.5%	0.5%
Chinese	0.5%	0.5%	0.5%
I'd rather not say	2.1%	2.1%	2.1%
Indian	0.7%	0.7%	0.7%
Other Mixed/Multiple ethnic background	0.7%	0.7%	0.7%
Other White background	3.2%	3.2%	3.2%
Other ethnic group	0.4%	0.4%	0.4%
White and Black Caribbean	0.2%	0.2%	0.2%
White: English/Welsh/Scottish/Northern Irish/British	88.0%	88.0%	88.0%
White: Irish	2.8%	2.8%	2.8%
Broad ethnic group, 5 categories			
White	94.0%	94.0%	94.0%
Mixed	0.9%	0.9%	0.9%
Asian	1.6%	1.6%	1.6%
Black	1.1%	1.1%	1.1%
Other	0.4%	0.4%	0.4%
Unknown	2.1%	2.1%	2.1%
Disability status			
Don't know or would rather not say	1.8%	1.8%	1.8%
No	85.9%	85.9%	85.9%

Yes, limited a little	11.5%	11.5%	11.5%
Yes, limited a lot	0.9%	0.9%	0.9%
Participation category			
Runner or walker and volunteer	14.3%	14.3%	14.3%
Runner or walker only	85.7%	85.7%	85.7%
Activity status			
Less than once per week	0.0%	2.8%	1.5%
About once per week	12.8%	10.8%	11.8%
About twice per week	21.5%	18.1%	19.7%
About three times per week	34.4%	29.9%	32.0%
Four or more times per week	31.2%	38.4%	35.0%
IPAQ category			
Low	27.9%	25.1%	26.6%
Moderate	63.7%	57.7%	61.0%
Vigorous	8.4%	17.1%	12.4%
Activity category at registration			
Less than once per week	9.0%	9.0%	9.0%
About once per week	13.5%	13.5%	13.5%
About twice per week	24.1%	24.1%	24.1%
About three times per week	29.9%	29.9%	29.9%
Four or more times per week	22.3%	22.3%	22.3%
Rather not say / don't know	1.2%	1.2%	1.2%
Club status			
Attached	10.2%	10.2%	10.2%
Unattached	89.8%	89.8%	89.8%
Deprivation Quartile			
1 - Most Deprived Quartile	9.9%	9.9%	9.9%
2nd Most Deprived Quartile	22.3%	22.3%	22.3%
3rd Most Deprived Quartile	32.0%	32.0%	32.0%
4 - Least Deprived Quartile	35.8%	35.8%	35.8%

Table A2. Demographics of the cross-sectional survey

parkrun participation	Registered but not yet participated	Runner or walker and volunteer	Runner or walker only	Volunteer only	Total Observations
Sample Size	1786	21928	38071	681	62466
Life satisfaction & happiness (1-10- raw)	7.12 (1786)	7.87 (21928)	7.70 (38071)	7.65 (681)	7.74 (62466)
Happiness (yesterday, 1-10 raw)	7.02 (1786)	7.66 (21928)	7.45 (38071)	7.55 (681)	7.51 (62466)
single physical activity question (1-7)	3.27 (1782)	3.79 (21914)	3.48 (38052)	3.16 (681)	3.58 (62429)
EQ5D - Mobility (No.)	1.16 (1781)	1.10 (21549)	1.08 (37563)	1.46 (666)	1.09 (61559)
EQ5D - Self care (No.)	1.04 (1774)	1.02 (21486)	1.02 (37409)	1.10 (662)	1.02 (61331)
EQ5D - Usual activities (No.)	1.18 (1770)	1.12 (21483)	1.10 (37374)	1.41 (661)	1.11 (61288)
EQ5D - Pain / discomfort (No.)	1.59 (1768)	1.49 (21464)	1.44 (37338)	1.80 (661)	1.46 (61231)
EQ5D - Anxiety / depression (No.)	1.65 (1762)	1.46 (21441)	1.50 (37310)	1.50 (661)	1.49 (61174)
Sum of EQ5D scores	6.61 (1762)	6.18 (21423)	6.12 (37298)	7.26 (661)	6.17 (61144)
EQ5D - VAS Scale (1-100)	77.42 (1713)	81.84 (20888)	80.53 (36398)	76.46 (647)	80.86 (59646)
No. of conditions	0.38 (1767)	0.20 (21805)	0.17 (37892)	0.72 (671)	0.19 (62135)
Work days lost to sickness (12 months)	4.35 (1481)	3.81 (17771)	3.40 (31774)	10.76 (463)	3.64 (51489)
Contact with GP (times, 12 months)	2.24 (1690)	1.91 (20589)	1.86 (35811)	3.21 (632)	1.90 (58722)
Contact with any other NHS medical specialists (times, 12 months)	1.59 (1609)	1.57 (19128)	1.34 (33490)	3.20 (586)	1.45 (54813)
Contact with any medical specialist outside of the NHS (times, 12 months)	0.75 (1515)	5.87 (17398)	3.54 (30920)	1.65 (513)	4.24 (50346)
Admitted to hospital (times, 12 months)	0.15 (1529)	0.13 (17257)	0.12 (30995)	0.26 (523)	0.13 (50304)
Been to A&E (times, 12 months)	0.20 (1521)	0.20 (17276)	0.16 (30934)	0.43 (526)	0.18 (50257)
Been an inpatient (days, 12 months)	0.70 (1528)	0.31 (17120)	0.25 (30789)	1.10 (522)	0.29 (49959)
WES - Optimism (No.)	3.61 (1698)	3.78 (20899)	3.76 (36116)	3.69 (637)	3.76 (59350)
WES - Useful (No.)	3.71 (1695)	3.89 (20867)	3.86 (36042)	3.86 (638)	3.87 (59242)
WES - Relaxed (No.)	3.41 (1698)	3.61 (20889)	3.53 (36074)	3.58 (637)	3.55 (59298)
WES - Problems (No.)	3.78 (1698)	3.92 (20912)	3.87 (36111)	3.94 (638)	3.89 (59359)
WES - Thinking (No.)	3.97 (1692)	4.11 (20884)	4.06 (36064)	4.10 (636)	4.08 (59276)
WES - Closeness (No.)	3.68 (1695)	3.84 (20886)	3.82 (36063)	3.83 (637)	3.82 (59281)
WES - Mind (No.)	4.19 (1695)	4.26 (20906)	4.22 (36101)	4.28 (637)	4.24 (59339)
WES Score (No.) Only those who answer all questions	26.35 (1679)	27.43 (20654)	27.12 (35686)	27.27 (628)	27.21 (58647)
Vigorous activity (METs per week)	1159 (1786)	1366 (21928)	1152 (38071)	925.79 (681)	1225 (62466)
Moderate activity (METs per week)	597.32 (1786)	569.01 (21928)	513.16 (38071)	515.92 (681)	535.20 (62466)
Walk activity (METs per week)	941.46 (1786)	869.94 (21928)	828.75 (38071)	953.86 (681)	847.80 (62466)
Total activity (METs per week)	2698 (1786)	2805 (21928)	2494 (38071)	2396 (681)	2608 (62466)
Years Registered: calculated	1.91 (1068)	4.17 (17952)	2.48 (28749)	3.00 (482)	3.10 (48251)
Total no. runs (if matched): calculation)	0.63 (1786)	80.43 (17833)	24.03 (27875)	31.47 (248)	44.26 (47742)
average number of runs per year since registration	0.32 (1068)	19.90 (17833)	13.61 (27875)	6.90 (248)	15.66 (47024)

Runs 2018 (matched data from Mike)	0.05 (1786)	18.05 (17833)	8.50 (27875)	3.24 (248)	11.72 (47742)
Volunteer occasions (raw data includes false zeros)	0.05 (1786)	20.17 (17833)	0.40 (27875)	56.44 (248)	8.06 (47742)
Volunteering occasions per year	0.03 (1068)	5.05 (17879)	0.14 (27884)	18.22 (377)	2.14 (47208)
End Date + Time	21493 (1786)	21493 (21928)	21492 (38071)	21494 (681)	21492 (62466)
AverageTime (from matched data) (mins)	32.63 (150)	29.94 (17833)	30.26 (27875)	36.30 (248)	30.17 (46106)
PB (from matched data) (mins)	31.68 (150)	26.45 (17833)	28.17 (27875)	32.98 (248)	27.55 (46106)
IMD From postcode, returned with matched data	17233 (1041)	18617 (17787)	18522 (28366)	17146 (481)	18515 (47675)
IMD% From postcode, returned with matched data	0.59 (1041)	0.64 (17787)	0.63 (28366)	0.62 (481)	0.64 (47675)
IMD quartile	2.85 (1041)	3.04 (17787)	2.99 (28366)	2.95 (481)	3.00 (47675)
Age (matched by Mike)	46.93 (1762)	50.72 (21794)	46.37 (37823)	53.81 (680)	48.00 (62059)
Gender (from matched data)					
F	56.6%	52.6%	51.1%	58.1%	51.8%
M	43.4%	47.4%	48.9%	41.9%	48.2%
Employment status					
Employed, but currently on maternity/paternity leave	0.7%	0.4%	0.5%	0.7%	0.4%
Employed, but currently on sick leave	0.3%	0.2%	0.2%	1.0%	0.3%
Full-time paid employment	54.1%	51.5%	56.4%	33.5%	54.4%
Fully retired	11.5%	15.2%	10.5%	30.7%	12.4%
Home maker	2.4%	1.5%	1.6%	1.9%	1.6%
I'd rather not say	0.9%	0.4%	0.4%	0.7%	0.4%
Other	1.7%	1.1%	1.0%	1.8%	1.1%
Part-time paid employment	10.2%	14.5%	13.2%	14.2%	13.6%
Retired, but still in paid employment	2.1%	2.8%	2.0%	2.2%	2.3%
Self-employed	9.6%	9.4%	9.3%	8.9%	9.3%
Student	4.3%	1.8%	3.8%	3.0%	3.1%
Unemployed and not working	2.2%	1.2%	1.2%	1.3%	1.2%
Ethnicity					
African	0.3%	0.1%	0.2%	0.0%	0.1%
Any other ethnic group, please describe	0.3%	0.1%	0.2%	0.1%	0.2%
Arab	0.1%	0.0%	0.0%	0.3%	0.0%
Bangladeshi	0.0%	0.0%	0.1%	0.1%	0.0%
Caribbean	0.5%	0.1%	0.2%	0.3%	0.2%
Chinese	0.3%	0.2%	0.2%	0.1%	0.2%
I'd rather not say	0.9%	0.9%	0.7%	1.5%	0.8%
Indian	0.9%	0.5%	0.7%	0.1%	0.6%
Other Asian background	0.6%	0.2%	0.2%	0.0%	0.2%
Other Black/African/Caribbean background	0.2%	0.1%	0.1%	0.1%	0.1%
Other Mixed/Multiple ethnic background	0.7%	0.4%	0.5%	0.1%	0.4%

Other White background	5.2%	3.4%	4.4%	3.6%	4.0%
Pakistani	0.2%	0.1%	0.2%	0.0%	0.1%
White and Asian	0.5%	0.3%	0.4%	0.1%	0.4%
White and Black African	0.2%	0.1%	0.1%	0.0%	0.1%
White and Black Caribbean	0.4%	0.2%	0.2%	0.1%	0.2%
White: English/Welsh/Scottish/Northern Irish/British	86.4%	90.8%	88.8%	91.1%	89.5%
White: Irish	2.3%	2.5%	2.6%	2.1%	2.6%
White: Irish Traveller	0.0%	0.0%	0.1%	0.0%	0.0%
Broad ethnic group, 5 categories					
White	91.9%	95.9%	95.0%	96.0%	95.2%
Mixed	1.8%	1.0%	1.3%	0.4%	1.2%
Asian	1.8%	0.9%	1.4%	0.4%	1.2%
Black	1.0%	0.3%	0.5%	0.4%	0.5%
Other	0.4%	0.2%	0.2%	0.4%	0.2%
Unknown	3.1%	1.7%	1.7%	2.2%	1.7%
Health condition, disability or illness					
Don't know or would rather not say	1.1%	0.6%	0.5%	1.5%	0.5%
No	80.9%	88.9%	90.6%	62.7%	89.4%
Yes, limited a little	15.3%	9.7%	8.2%	26.3%	9.1%
Yes, limited a lot	2.7%	0.9%	0.8%	9.5%	1.0%
Pregnant last 12 months (Yes / No)					
No	97.7%	98.5%	98.0%	97.2%	98.2%
Would rather not say	0.6%	0.2%	0.2%	0.1%	0.2%
Yes	1.7%	1.3%	1.7%	2.7%	1.6%
parkrun participation					
Registered but not yet participated	100.0%	0.0%	0.0%	0.0%	2.9%
Runner or walker and volunteer	0.0%	100.0%	0.0%	0.0%	35.1%
Runner or walker only	0.0%	0.0%	100.0%	0.0%	60.9%
Volunteer only	0.0%	0.0%	0.0%	100.0%	1.1%
parkrun physical activity question (last month 0-4+)					
Less than once per week	9.1%	1.8%	3.6%	13.4%	3.3%
About once per week	9.5%	4.9%	7.8%	12.1%	6.8%
About twice per week	17.0%	13.9%	18.1%	16.1%	16.6%
About three times per week	26.4%	31.0%	30.6%	23.2%	30.5%
Four or more times per week	38.0%	48.4%	39.9%	35.3%	42.8%
IPAQ Category (Text)					
High	58.3%	68.4%	61.6%	51.8%	63.9%
Low	11.6%	6.2%	8.1%	9.8%	7.6%
Moderate	30.0%	25.4%	30.3%	38.4%	28.6%
Pre-parkrun exercise frequency (from					

matched data)					
Less than once per week	6.9%	4.0%	5.1%	7.7%	4.8%
About once per week	10.6%	9.4%	11.2%	11.4%	10.5%
About twice per week	19.6%	19.3%	21.8%	20.1%	20.8%
About three times per week	32.3%	30.0%	31.5%	25.9%	30.9%
Four or more times per week	26.8%	24.6%	24.6%	22.4%	24.6%
Rather not say / don't know	0.7%	0.6%	0.9%	4.4%	0.8%
Unset	3.2%	12.1%	4.9%	8.1%	7.6%
IMD quartile					
1	12.4%	8.6%	10.1%	12.1%	9.6%
2	24.3%	20.0%	20.6%	18.1%	20.4%
3	29.2%	30.6%	29.6%	32.8%	30.0%
4	34.1%	40.8%	39.7%	37.0%	40.0%

List of full analysis results

Longitudinal survey - descriptive statistics and full regression results

Longitudinal survey - regression results with health control

Cross-sectional survey - descriptive statistics and full regression results

Cross-sectional survey - regression results with health control